

VHS: Virtuous data accesses in High-capacity Storage systems

VHS– Common 2025 ANR call – JCJC (Jeunes chercheurs et jeunes chercheuses) – Duration: 42 months
Axe E.03 : Sciences et génie du logiciel – Réseaux de communication multi-usages, infrastructures numériques

CONTEXT, POSITIONING AND OBJECTIVES

Context. Magnetic tapes have been used to store computer data since the 1950s, so the layman now often considers it as an outdated technology. However, tape storage is still and will remain essential in many fields such as academic research (e.g., high-energy physics projects), international organisations (e.g., ECMWF) or cloud companies [1] as it is a technology used for all the cold-storage solutions. This dependency on tapes has motivated industrial efforts in technology improvements, resulting in much faster data density progression on tape rather than on disk. After recent breakthroughs in materials used, tape capacity is even expected to witness a massive leap in the coming years, increasing from 20TB to several hundreds of TB per tape [2]. This evolution will amplify the main benefits of tape storage, i.e., considerably lower costs and energy consumption, along with higher security with air gap protection when tapes are stored on the shelves.

However, tapes have often been primarily considered for archiving cold data, because of their main drawback: it takes around a minute to mount a tape from its shelf into a drive and position the reading head before starting reading data. This explains the current lack of academic effort to optimize relatively frequent data accesses. Nevertheless, more and more research projects require to handle tremendous volumes of data, which are not only destined to be archived but also regularly accessed for scientific analysis. Budget constraints impose the usage of tape storage, despite their inherent latency. Optimizing tape data access therefore becomes more and more significant, and not limited to improving archive retrieval. Examples of such research projects are numerous and include the LHC at CERN (1EB currently stored on tape, tens of PB produced per year), LSST (7 PB produced per year), SKA (300PB produced per year) or Euclid (150PB produced).

Positioning. Hierarchical Storage Management (HSM) is a “venerable” paradigm, used in HPC since the late 80s. It strongly evolves in the Exascale era as new challenges appear and new technologies become available. In particular, the storage stack changes from a tape-to-disk hierarchy to more complex hierarchies involving three, four or even five levels, including different kinds of Solid-State Drives (SSDs) or rotative hard disk drives.

Those new hierarchies aim to benefit from advantages of each technology: storage capacity offered by tapes, which are slow, speed and high throughputs offered by rather small SSDs. In this context, optimizing the throughput when accessing each piece of hardware is very important.

Contrarily to disks, few works have focused on the tape technology itself, hence optimizing the data accesses at the tape level. Some studies [3] at CERN have focused on benchmarking timings when moving along the tape, to derive a cost model for the positioning times. However, such a work seems difficult to generalize in any hardware and software setup, and suffers from a lack of knowledge on hardware behavior. Other approaches [4] have studied hardware issues on tape, but are outdated. Some theoretical studies [5] have proposed algorithms to schedule reading operations on simple tape models. Such strategies are hardly adaptable in a real setup due to the complex mechanical behavior of tapes for which we cannot provide cost models to instantiate correctly algorithm parameters.

Some projects have focused on tape storage, such as The Data Carousel Project [6] at CERN. Its objective is to stream tape data to disks so that they can be processed in a pipeline without waiting for all data to be transferred. However, the solutions are very specific to the data access pattern, and do not focus on improving the tape system performance itself. IO-SEA [7] is a EuroHPC funded project that also focuses on HSM paradigm. It aims at building an IO (Input/Output) software stack for

Exascale. IO-SEA intensively makes use of tapes, interfacing it with classical disk filesystem. IO-SEA focuses on the software stack itself, by introducing new paradigms such as ephemeral services, and makes use of object store. IO-SEA uses tapes as a brick of the hierarchy to derive a complete IO stack, but does not provide performance enhancement for tape itself. VHS focuses on a smarter use of tapes, making it possible to introduce new directions for data access optimization. Thus, IO-SEA would benefit from those results by making a better use of capacitive but slow devices such as tapes.

All the aforementioned projects focus uses tapes as a brick of the HSM, by providing high-level optimizations that do not take into account the interactions between the tape drive's firmware and the tape drive support provided in the operating system. The VHS project will act as a lower-level, by smartly reorganizing IO operations, leading to a better usage of the available IO bandwidth to tapes and an higher aggregated IO throughput on the overall HSM.

Magnetic tape is a very sophisticated storage, with a long-term roadmap [2], that did not receive the same attention as disk storage had in the computing community. Our ambition in the VHS project is to focus on the interactions between the operating system and the tape storage firmware itself, in order to perform more efficient operations on tapes. To the best of our knowledge, this approach is novel as it does not only uses tape systems as they are, but it tries to optimize its usage in the HSM. This approach is complementary to the aforementioned initiatives.

Objectives. The objective of the VHS project is to increase IO throughput at the tape level, so that performance of the global HSM is improved.

The general idea of the VHS project is to propose new interactions between resource management and tape systems. Using filesystem on tapes such as LTFS [8], we plan to design novel data placement strategies that will propose efficient data accesses on tape by optimizing its operational cost. Such strategies will allow the operating system to decide **when to make use of tapes** and **at what cost**, using **efficient operations** on tapes, thus enhancing the overall performance of the tape storage. In this objective, we propose a bi-directional yet complementary approach. The former investigates performance improvements at the tape level itself, the latter at the level of the interactions between the tape system and the HSM.

1. Improving operations on tape based on firmware operations The first direction of the VHS project is to focus on the tape component itself, to find levers in order to optimize data access operations. Contrary to disks, very few works in the literature have studied the impact of the mechanical processes on the performance of read/write operations. We plan to investigate if optimizations on the internal processes of tapes can be done to accelerate data access on tapes, using the functionalities of the firmware. The literature lacks understanding of internal tape behavior, hence limiting the possibilities to use the firmware functionalities to perform local optimizations for data accesses. To achieve this step, we plan to perform extensive benchmarking on high performance tape drives, associated with fine-grain performance evaluation based on extracted performance metrics and firmware interactions. Partners of the consortium managing large tape systems will be an asset to achieve these objectives.

2. Improving tape integration in HSM software While our first direction primary focus is tape internal behavior, we aim in this direction to study the support software that connects tape drive's firmware and the operating system. This software is the central piece in our objective of improving IO throughput of the overall HSM. We want to obtain permissiveness at the tape level of the HSM, by leveraging knowledge of IO requirements from the OS. We envision in that direction to derive **efficient requests to the tape** from the software connecting it to the HSM. For instance, prioritizing operations on a tape that is unwound at a given point should perform better than unwinding the tape for each of those operations. For writing data on tapes, the problem of determining how to pack data on tapes based on properties (user, project, etc) in order to improve future reading operations is also

challenging. Solutions to those aspects require both characterization of efficient operations on tape, but also a view on the IO requests on it, given by the OS. This direction of the project will have strong interactions with the global resource management system.

PROJECT ORGANIZATION

The Principal Investigator of the VHS project is Valentin Honoré. He will devote 80% of his research time to the project. Valentin Honoré is an Associate Professor at the Ecole Nationale Supérieure d'Informatique pour l'Industrie et les Entreprises (ENSIIE). He is a member of the Inria/Samovar Benagil team, which focuses on efficient operating systems for HPC/Cloud. He received his PhD from the University of Bordeaux in 2020. His research focuses on scheduling algorithms and performance models in distributed systems. He has been working on tape-related problems since 2021 [5].

In addition to the principal investigator, the VHS project will involve other members from the Benagil Team with a strong expertise in performance analysis and runtime system (Prof. François Trahay). Dr. Bertrand Simon from the Centre de Calcul de l'IN2P3 (CC-IN2P3) will bring an expertise on algorithm design and knowledge of tape usage in this computing center. The consortium will also benefit from the expertise of the Commissariat à l'Énergie Atomique (CEA) in the maintenance and operation of large-scale tape storage system. Dr. Philippe Deniel, an expert on tape storage and member of the Storage Laboratory of CEA, will closely collaborate with the academic partners at each step of the project, with the objective to build efficient solutions, adapted to a production tape system. All these skills will be combined to achieve the objectives of the VHS project.

Funding. For this project, we ask for 216k€ including overheads: 1 PhD student ($\sim 148\text{k€}$), hardware (including high performance tape drive, $\sim 35\text{k€}$), two annual trips to conferences for two researchers ($\sim 10\text{k€}$), the funding for organizing workshops at different steps of the project ($\sim 8\text{k€}$).

IMPACT AND METHODOLOGY

Methodology. In the VHS project, we target fully open-source solutions. As we expect to impact not only academic actors but also other tape users, open-source solutions are at the core of the proposal of this project. The CEA has dedicated many efforts in the development of open-source software dedicated to tape storage systems. We plan to rely on Phobos [9], a tool designed to manage large volumes of data for various kinds of storage technologies from disks to tapes. Phobos uses the LTFS filesystem [8] to interact with the tapes. We plan to integrate our software solutions with Phobos as it is already capable of performing operations both on disks and tapes. The expertise of the partners from the CEA in the development of Phobos will be a strong asset to achieve this objective.

Performance evaluation on a production tape storage system is challenging, due to security restrictions and limited action range. We intend to first develop and benchmark solutions on a local setup including hardware for which we ask funding in this project. Even though the setup does not benefit of the scale of a production system, the hardware and software environment will be very similar to the one we could find in the installations of partners such as CEA. Hence, it will be sufficient to achieve the objectives of this project and demonstrate the practicality of our solutions.

Impact. One major impact of this project will be to provide better usage of tape storage by refining the interactions between HSM layers and tape system. Tape technologies and some disk technologies tend to converge in some ways. Servo-tracks on IBM Jaguar Tapes and later in LTO is an example [10]. Optimisations rely on a fine knowledge about disk behavior from the firmware, the operating system and some applications. We believe a similar approach is to be used with tapes, with benefits to all communities using this technology. With the important pressure on HSM due to the amount of data to be managed, improvements at one level of the hierarchy is beneficial to all levels of the HSM.

Recently, large-scale initiatives on Exascale Computing such as PEPR NumPEX have been initiated in Europe, with strong constraints on the sustainability of such systems. NumPEX project focuses on the full stack to deploy supercomputers. This project does not yet consider tapes, and focuses on classical storage medium that are used by applications (DRAM, disks etc). We believe that the VHS project is complementary to this initiative, as disks are too costly to store all data from exascale machines. Thus, we hope this project will trigger or enforce collaborations with the computing communities to provide more efficient storage systems with tapes. We also intent to contact other institutions working on tape storage (e.g. CERN), and organize joint workshops to discuss related issues and approaches. The VHS project addresses the "Défi 7 > Axe E.3 : Sciences et génie du logiciel - Réseaux de communication multi-usages, infrastructures numériques" challenge. Indeed, we intend to increase the performance of tape storage system that are widely used in computation-intensive facilities. Such objectives will address the need of sustainability of hardware resources and a dynamic management of data at scale. In addition, the VHS project also addresses several "*Technical Research Priorities*" from the 2022 ETP4HPC Strategic Research Agenda [11], in particular priority #3 *System Software and Management* and priority #5 *IO and Storage*.

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